

Trainings ▾

General Information

This section includes recommended glycol concentration, recommended supplemental coolant additives (SCA) concentration for Cummins® coolants, and replacement limits for coolants. This section also includes action to take if coolant does **not** fall within the recommendations and limits.

SCA limits are specific to coolant formulation. This section outlines limits for Fleetguard® coolants **only**. For other coolant brands, consult coolant manufacturer for recommendations.

Glycol recommendations and coolant replacement limits are independent of coolant formulation and can be applied to both Fleetguard® and other coolant brands.

Glycol Concentration

Cummins Inc. requires a glycol/antifreeze concentration between 40 to 60 percent by volume unless different glycol/antifreeze concentrations are specifically approved. In arctic conditions (-32 to -54°C [-25 to - 65°F]) as defined in Service Bulletin, Operation of Diesel Engines in Cold Climates, Bulletin 3379009 (/qs3/pubsys2/xml/en/bulletin/3379009.html), Cummins Inc. recommends the use of a 60 percent ethylene glycol coolant for arctic specifications. Ethylene glycol concentration is **not** to exceed 68 percent in arctic conditions.

Table 1 - Glycol Concentration

Concentration	Out of Range Actions
40 to 60 Percent	Drain a portion of coolant and replace with 50/50 coolant to get back in the 40 to 60 percent glycol range. Under-concentration of glycol occurs when filling or topping off with straight water or over-diluted coolant. Over-concentration of glycol occurs in high heat situations where there is some water evaporation from the cooling system or when filling or topping off with improperly diluted coolant.

SCA Concentration for Fleetguard® Coolants

SCA units are a measure of how much nitrite and/or molybdate are in a coolant. Table 2 provides guidelines on maintaining proper SCA concentrations when using standard or extended service intervals. Fleetguard® recommends targeting an SCA concentration of 0.6 units/L [2.5 units/U.S. gal] in coolant solutions of 40 to 60 percent glycol. SCA concentration are to be determined by test strip or lab test prior to adding SCA.

Table 2 - Guidelines for SCA Addition

SCA Concentration	Recommendation
Less than 0.3 units/L [1.2 units/U.S. gal]	Add 0.15 L [5 oz.] of liquid SCA per 3.8 L [1 U.S. gal] of coolant. Install the correct chemical coolant filter, if applicable. For large cooling systems, it may be necessary to use liquid SCA in conjunction with a chemical filter to replenish SCA. All diesel engines with wet cylinder liners are subject to cylinder liner pitting if the cooling system is not correctly maintained. Concentration below 0.3 units/L [1.2 units/U.S. gal] of SCA results in cylinder liner cavitation and possible engine malfunction.
0.3 units/L [1.2 units/U.S. gal] to 0.8 units/L [3.0 units/U.S. gal]	Install the correct chemical coolant filter, if the engine is equipped with a coolant filter head, or add the correct liquid SCA and install a chemical-free coolant filter.
0.8 units/L [3.0 units/U.S. gal] to 1.3 units/L [5.0 units/U.S. gal]	Install the correct chemical free coolant filter, if applicable. Do not add liquid SCA. Test the SCA level at each oil change until the SCA drops below 0.8 units/L [3.0 units/U.S. gal]. Maintaining SCA concentration between 0.8 to 1.3 units/L [3.0 to 5.0 units/U.S. gal] does not offer any additional benefit over the suggested concentration range, but generally will not cause issues.

Table 2 - Guidelines for SCA Addition

SCA Concentration	Recommendation
Greater than 1.3 units/L [5.0 units/U.S. gal]	Install the correct chemical-free coolant filter, if applicable. Do not add liquid SCA. Test the SCA level at each oil change until the SCA drops below 0.8 units/L [3.0 units/U.S. gal]. Drain and replace the coolant if overheating occurs or if cooling system corrosion is detected by visual changes to coolant's appearance or by laboratory testing. Otherwise, letting the SCA deplete naturally is generally safe. Over-concentration of additives or the use of high-silicate antifreeze can result in silicate gelation or water pump seal leakage as well as aluminum and solder corrosion.

In arctic conditions, -32 to -54°C [-25 to - 65°F], as defined in Service Bulletin, Operation of Diesel Engines in Cold Climates, Bulletin 3379009 (/qs3/pubsys2/xml/en/bulletin/3379009.html), maintaining SCA concentration between 0.3 and 0.8 units/L [1.2 to 3.0 units/U.S. gal] is critical. Overconcentration of SCA about 0.8 units/L (3.0 units/U.S. gal) **must** be avoided because of limited solubility of coolant additives at low temperatures and higher concentrations of glycol.

Fleetguard® sells two SCA products that are marketed as diesel coolant additive (DCA) -2 and DCA-4. See table 3 below for recommendations of which SCA product to use with various coolants.

Note : Organic acid technology (OAT)/Extended Life coolants use high levels of salts of organic acids for cylinder liner pitting protection and do **not** use nitrite and or molybdate. Therefore, these coolants do not have SCA numbers. Additionally, OAT additives deplete much slower than conventional additives and as a result OAT coolants do not require addition of SCA products.

Table 3 - Fleetguard® SCA Products

SCA Product	Description	Fleetguard® Coolants
DCA-2	Recommended for use with coolants containing nitrite and borate	Fleetcool™ Compleat™
DCA-4	Recommended for use with coolants containing nitrite, molybdate, and phosphate	ES Compleat™ Fleetcool™ EX PG Plus™

Coolant Replacement Requirements

Table 4 below lists the condemning limits for various coolant contaminants.

Table 4 - Coolant Replacement Limits		
Contaminant	Allowable Level	Out of Range Actions
Sulfate (SO ₄)	1500 ppm, maximum	Drain and replace with new coolant
Chloride (Cl ⁻)+, Flouride (F ⁻)+, Bromide (Br ⁻)	100 ppm, combined maximum	
Glycolate	3000 ppm, maximum	
pH	6.5 minimum / 10.5 maximum Note: Some Cummins® Engineering Standards approved coolants start at pH levels of 10.5 and will decrease naturally with time.	
Oil or fuel contamination	Coolant must be free of these contaminants	Drain, clean, and replace with new coolant. Refer to Procedure 379-009 in Section 5. (/qs3/pubsys2/xml/en/proce dures/00/00-379-009.html)
Grease, solder bloom, silica gel, rust, or scaling	Coolant must be free of these contaminants	